

# Quantitative Aptitude: Percentage, Ratio and Average

संख्यात्मक अभियोग्यता: प्रतिशत, अनुपात और औसत

How this topic appears in the CRT

Percentage, ratio and average are the three arithmetic tools behind most quant questions in UPSC Recruitment Tests, and they come dressed as school data.

Official syllabus topic (iii) is "Reasoning ability and Quantitative Aptitude". The level is that of the General Ability Test in UPSC Recruitment Tests, well below CSAT: one- or two-step arithmetic without a calculator. GAT-pattern papers suggest roughly **15-20% of the questions** come from topic (iii), and percentage, ratio and average are the base for profit and loss, interest, mixtures, speed and data interpretation.

Expect the numbers to be about enrolment, marks, attendance, fees, grants and teacher-student ratios. A wrong answer costs **one-third of the question's marks**, so a blind guess among four options has zero expected value. Work the question, check the base, and move on if it takes more than about 90 seconds.

**Percentage: the basics**

Per cent means "per hundred"; every percentage question is a fraction with denominator 100.

- $x\%$  of  $y = x \times y / 100$ . So  $45\%$  of  $1,200 = 0.45 \times 1,200 = \mathbf{540}$ .
- "What per cent is  $a$  of  $b$ "  $= a / b \times 100$ . So  $324$  out of  $450 = 324/450 \times 100 = \mathbf{72\%}$ .
- Fraction to per cent: multiply by  $100$ . Per cent to fraction: divide by  $100$  and simplify.
- Per cent to decimal: move the point two places left ( $12.5\% = 0.125$ ).

Memorise the fraction-per cent table. It turns most questions into mental arithmetic.

| Fraction | %                | Fraction | %                |
|----------|------------------|----------|------------------|
| $1/2$    | 50               | $1/8$    | 12.5             |
| $1/3$    | $33 \frac{1}{3}$ | $1/9$    | $11 \frac{1}{9}$ |
| $1/4$    | 25               | $1/11$   | $9 \frac{1}{11}$ |
| $1/5$    | 20               | $1/12$   | $8 \frac{1}{3}$  |
| $1/6$    | $16 \frac{2}{3}$ | $1/16$   | 6.25             |
| $1/7$    | $14 \frac{2}{7}$ | $1/20$   | 5                |

Fast tricks:  $10\%$  is one-tenth,  $5\%$  is half of that,  $1\%$  is one-hundredth. To find  $15\%$  of  $840$ :  $10\% = 84$ ,  $5\% = 42$ , total **126**. Also  $a\%$  of  $b = b\%$  of  $a$ , so  $8\%$  of  $25 = 25\%$  of  $8 = \mathbf{2}$ .

## Percentage change and the base trap

Percentage change is always measured on the original (earlier) value; changing the base is the most common wrong option.

Percentage change  $= (\text{new} - \text{old}) / \text{old} \times 100$ . Enrolment rising from  $800$  to  $920$  is a change of  $120$  on a base of

800, so **+15%**. The wrong option 13.04% comes from dividing by 920.

New value = old  $\times (100 \pm x)/100$ . A 12% rise multiplies by 1.12; a 12% fall multiplies by 0.88. The multiplier approach is faster than computing the change and adding it.

"More than" and "less than" use different bases:

- If A is  $x\%$  more than B, then B is  **$100x/(100 + x) \%$**  less than A.
- If A is  $x\%$  less than B, then B is  **$100x/(100 - x) \%$**  more than A.

| A compared with B | B compared with A |
|-------------------|-------------------|
| 20% more          | 16 2/3% less      |
| 25% more          | 20% less          |
| 50% more          | 33 1/3% less      |
| 20% less          | 25% more          |
| 25% less          | 33 1/3% more      |

Worked: A's marks are 25% less than B's. Take  $B = 100$ ,  $A = 75$ . B exceeds A by 25 on a base of 75 = **33 1/3%** more, not 25%.

Percentage points versus per cent: if the pass percentage goes from 60% to 72%, it rose by **12 percentage points** but by **20 per cent** (12/60). UPSC options exploit this difference.

## Successive changes, expenditure and growth

**Two percentage changes never simply add; use the net-change formula.**

Successive changes of  $x\%$  and  $y\%$  (with signs) give net  
 $= x + y + xy/100$ .

| Changes        | Net  |
|----------------|------|
| +20% then +25% | +50% |
| +10% then -10% | -1%  |
| -10% then -10% | -19% |
| +20% then +20% | +44% |
| -20% then -10% | -28% |

Worked: a fee is raised 20% and then 25%. Net =  $20 + 25 + (20 \times 25)/100 = \mathbf{50\%}$ , not 45%. Two discounts of 10% and 10% equal a single discount of **19%**, not 20%.

Undoing a change: a number increased by 25% must be reduced by **20%** to return to the original; a number reduced by 20% must be increased by 25%. The same formula covers price and consumption: if the price rises  $x\%$ , cut consumption by  $\mathbf{100x/(100 + x) \%}$  to keep spending constant. A 25% rise means a 20% cut; a 20% rise means a  $16 \frac{2}{3}\%$  cut.

Growth over  $n$  periods at  $r\%$  per period: final =  $P(1 + r/100)^n$ . Enrolment of 1,000 growing 10% a year reaches **1,210** in two years (not 1,200). Depreciation uses  $(1 - r/100)^n$ .

## Percentages in school data

The CRT frames these as marks, attendance, fees and the exam's own negative marking.

- Marks percentage = marks obtained / maximum marks  $\times 100$ . Converting a score of 54 out of 80 to a 100 scale gives **67.5%**.
- Attendance percentage = days present / working days  $\times 100$ . A student present 171 of 190 days has 90% attendance.
- Pass percentage of a school = students passed / students appeared  $\times 100$ , never divided by enrolment.
- Negative marking: with marks  $m$  per question and a penalty of  $m/3$ , net score = (correct  $\times m$ ) - (wrong  $\times m/3$ ). Three wrong answers cancel one correct answer.

Worked: a 150-question, 300-mark test (2 marks each) with one-third penalty. A candidate attempts 120, gets 90 correct and 30 wrong. Score =  $180 - 30 \times (2/3) = 180 - 20 = \mathbf{160}$ , which is **53 1/3%** of 300.

Worked: a school of 1,200 students has 45% girls. Boys = 55% of 1,200 = **660**. The trap option 540 is the number of girls.

## Ratio: the basics

A ratio compares two quantities of the same unit; it has no unit of its own and should always be in lowest terms.

- $a : b$  is read "a is to b";  $a$  is the antecedent,  $b$  the consequent. Multiplying or dividing both terms by

the same non-zero number leaves the ratio unchanged.

- Convert to the same unit first: 50 paise : Rs 2 = 50 : 200 = **1 : 4**.
- A ratio of three terms  $a : b : c$  assigns shares  $a/(a+b+c)$ ,  $b/(a+b+c)$ ,  $c/(a+b+c)$ .

Dividing a sum in a ratio: share = (part / sum of parts)  $\times$  total. A grant of Rs 36,000 split among library, sports and laboratory in 2 : 3 : 4 (9 parts, Rs 4,000 per part) gives Rs 8,000, **Rs 12,000** and Rs 16,000.

Combining ratios with a common term: make the common term equal. If  $a : b = 3 : 4$  and  $b : c = 2 : 5$ , write b as 4 in both:  $b : c = 4 : 10$ , so  $a : b : c = \mathbf{3 : 4 : 10}$  and  $a : c = 3 : 10$ . Shortcut for  $a : c$ : multiply the antecedents and the consequents,  $3 \times 2 : 4 \times 5 = 6 : 20 = 3 : 10$ .

Ratio-change questions: let the quantities be  $3x$  and  $5x$ . If adding 10 to each makes the ratio 5 : 7, then  $(3x + 10)/(5x + 10) = 5/7$ , so  $21x + 70 = 25x + 50$ ,  $x = 5$  and the numbers are **15 and 25**. Always check the answer back in the original ratio.

### Proportion and the special terms

**A proportion is an equality of two ratios; its vocabulary (mean, third, fourth proportional) is tested directly.**

$a : b :: c : d$  means  $a/b = c/d$ , so **ad = bc** (product of extremes = product of means). a and d are the extremes, b and c the means.

| Term                           | Definition                            | Example                    |
|--------------------------------|---------------------------------------|----------------------------|
| Fourth proportional to a, b, c | x with $a : b :: c : x$ , $x = bc/a$  | 2, 3, 8 gives <b>12</b>    |
| Third proportional to a, b     | x with $a : b :: b : x$ , $x = b^2/a$ | 4, 8 gives <b>16</b>       |
| Mean proportional between a, b | $\sqrt{ab}$                           | 4 and 9 gives <b>6</b>     |
| Duplicate ratio of a : b       | $a^2 : b^2$                           | 2 : 3 gives <b>4 : 9</b>   |
| Sub-duplicate ratio of a : b   | $\sqrt{a} : \sqrt{b}$                 | 16 : 25 gives <b>4 : 5</b> |

Direct proportion: more of one means more of the other (cost and quantity). Inverse proportion: more of one means less of the other (workers and days). The unitary method handles both: find the value for one unit, then scale.

Worked (inverse): 20 teachers check an answer-book bundle in 12 days. With 30 teachers, days =  $20 \times 12 / 30 = 8$ .

Teacher-student ratio: with 1,500 students and a ratio of 1 : 30 there are 50 teachers. After 10 more are posted, 60 teachers give  $1,500/60 = 25$ , so the ratio is **1 : 25**. Pupil-teacher ratio questions in school management papers also quote the RTE Schedule norms of **30 : 1 for Classes I-V** and **35 : 1 for Classes VI-VIII**.

Partnership: profit shares are in the ratio of capital  $\times$  time. Rs 20,000 for 12 months against Rs 30,000 for 8 months is  $240 : 240 = 1 : 1$ .

## Average: the basics

**Average (arithmetic mean) = sum of observations / number of observations; nearly every question reduces to finding the sum.**

- Sum = average  $\times$  count. Keep this line at the top of your rough work.
- Adding  $k$  to every observation raises the average by exactly  $k$ ; multiplying every observation by  $k$  multiplies the average by  $k$ .
- The average always lies between the smallest and largest value.
- Average of the first  $n$  natural numbers =  $(n + 1)/2$ ; the first 50 average 25.5, not 25.
- Average of consecutive numbers, consecutive even or odd numbers, or any arithmetic progression =  $(\text{first} + \text{last})/2$  = the middle term.

Worked: five unit tests average 68 (sum 340). A sixth test of 80 raises the sum to 420, so the new average =  $420/6 = 70$ .

## Adding, removing and replacing

**When one observation enters or leaves, work through the change in the total.**

- New member joins: new average =  $(\text{old sum} + \text{new value})/(n + 1)$ .
- Member leaves: new average =  $(\text{old sum} - \text{value})/(n - 1)$ .
- Replacement (count unchanged): new value = old value +  $n \times (\text{change in average})$ .



Worked: 30 students average 45 kg (sum 1,350). A 60-kg student leaves and another joins, and the average falls to 44.5 (sum 1,335). The newcomer weighs  $60 - 15 = \mathbf{45 \text{ kg}}$ . Using the formula:  $60 + 30 \times (-0.5) = 45$ .

Worked: 40 students average 62 (sum 2,480).

Excluding a score of 23 leaves  $2,457/39 = \mathbf{63}$ . Giving all 40 students 5 grace marks makes the average **67**.

Overlapping groups: the average of 11 numbers is 50 (sum 550). The first six average 49 (sum 294) and the last six average 52 (sum 312). The sixth number is counted twice, so it equals  $294 + 312 - 550 = \mathbf{56}$ .

Age questions: the average age of a group rises by exactly 1 each year because every member ages by 1. If a teacher joins and the average of 20 students rises from 14 to 16 years, the teacher's age is  $21 \times 16 - 20 \times 14 = 336 - 280 = \mathbf{56}$ .

### **Weighted average and combined groups**

**When groups of different sizes are combined, weight each average by its size; the simple mean of the averages is the trap.**

Combined average =  $(n_1 \times A_1 + n_2 \times A_2)/(n_1 + n_2)$ .

Section A has 40 students averaging 60 and Section B has 60 students averaging 70. Combined =  $(2,400 + 4,200)/100 = \mathbf{66}$ , not 65. The result is pulled towards the larger group.

Alligation gives the same answer backwards: the ratio of group sizes =  $(A_2 - \text{combined}) : (\text{combined} - A_1)$ . With 60 and 70 averaging 66, sizes are in the ratio  $(70 - 66) : (66 - 60) = 4 : 6 = \mathbf{2 : 3}$ .

Average speed for equal distances =  $\frac{2xy}{x+y}$ , never  $(x+y)/2$ . Going at 40 km/h and returning at 60 km/h averages **48 km/h**. For equal times the simple mean  $(x+y)/2$  is correct.

Mean, median and mode: the mean uses all values and is pulled by extremes; the median is the middle value when ordered; the mode is the most frequent value. Educational measurement questions overlap here, and the mean of a batch's marks is what "average" means in the quant paper.

### **A quick practice set**

**Five one-minute drills in CRT style; cover the answers and try them.**

1. Boys : girls in a class is 7 : 5 and there are 36 students. Girls =  $\frac{5}{12} \times 36 = \mathbf{15}$ .
2. A fee of Rs 500 is raised 10% a year for two years:  
 $500 \times 1.21 = \mathbf{Rs\ 605}$ .
3. 30% of a number is 96. Number =  $96/0.3 = \mathbf{320}$ .
4. An average of 8 numbers is 20; one number is removed and the average becomes 18. Removed number =  $160 - 126 = \mathbf{34}$ .
5. If  $36 : x :: x : 16$ , the mean proportional  $x = \sqrt{576} = \mathbf{24}$ .

### **Common traps**

- **Wrong base.** "20% more" does not reverse to "20% less". Percentage change uses the earlier value, never the later one.

- **Adding successive percentages.** +20% then +25% is +50%, not +45%; two 10% discounts are 19%, not 20%.
- **Percentage points versus per cent.** 60% to 72% is 12 points but a 20% rise.
- **Pass percentage base.** Divide by students who appeared, not by enrolment.
- **Count of the "other" group.** 45% girls asked as "how many boys" means use 55%.
- **Different units in a ratio.** Convert paise and rupees, minutes and hours, before forming the ratio.
- **Combining ratios.** Equalise the common term; do not cross-add.
- **Proportional vocabulary.** Third proportional to a, b is  $b^2/a$ ; mean proportional is  $\sqrt{ab}$ ; fourth proportional to a, b, c is  $bc/a$ .
- **Simple mean of averages.** Combined average must be weighted by group size.
- **Average speed.** Equal distances use  $2xy/(x + y)$ , not the simple mean.
- **First n natural numbers.** Average is  $(n + 1)/2$ , so 25.5 for  $n = 50$ .
- **Replacement direction.** New value = old value +  $n \times$  (new average - old average); watch the sign.
- **Negative marking.** Three wrong answers wipe out one correct; a blind guess has zero expected value.

### Remember for the exam

1.  $x\%$  of  $y = xy/100$ ;  $a\%$  of  $b = b\%$  of  $a$ .
2. Percentage change =  $(\text{new} - \text{old})/\text{old} \times 100$ .

3.  $x\%$  more means  $100x/(100 + x) \%$  less;  $x\%$  less means  $100x/(100 - x) \%$  more.
4. Successive changes: net =  $x + y + xy/100$ ;  $+10\%$  and  $-10\% = -1\%$ .
5. Price up  $x\%$ : cut consumption by  $100x/(100 + x) \%$ ;  $25\%$  up =  $20\%$  cut.
6. Growth:  $P(1 + r/100)^n$ ;  $10\%$  for 2 years =  $21\%$ .
7. Share in a ratio = (part/sum of parts)  $\times$  total.
8.  $a : b$  and  $b : c$  combine by equalising  $b$ ;  $a : c =$  product of antecedents : product of consequents.
9.  $a : b :: c : d$  means  $ad = bc$ ; mean proportional =  $\sqrt{ab}$ ; third proportional =  $b^2/a$ .
10. Duplicate ratio of  $a : b = a^2 : b^2$ .
11. Average = sum/count; sum = average  $\times$  count.
12. Average of first  $n$  natural numbers =  $(n + 1)/2$ ; of any AP = (first + last)/2.
13. Replacement: new value = old value +  $n \times$  (change in average).
14. Weighted average =  $(n_1A_1 + n_2A_2)/(n_1 + n_2)$ ; it leans towards the bigger group.
15. Average speed for equal distances =  $2xy/(x + y)$ ; 40 and 60 give 48.

### Quick facts

- Percentage change =  $(\text{new} - \text{old})/\text{old} \times 100$ ; the base is always the earlier value.
- If A is  $x\%$  more than B, B is  $100x/(100 + x)\%$  less than A:  $20\%$  more =  $16 \frac{2}{3}\%$  less;  $25\%$  less =  $33 \frac{1}{3}\%$  more.

- Successive changes: net =  $x + y + xy/100$ ; +20% then +25% = +50%; two 10% discounts = 19%.
- Price rises  $x\%$ : cut consumption by  $100x/(100 + x)\%$  to keep spending constant; 25% rise = 20% cut.
- Growth at  $r\%$  for  $n$  periods =  $P(1 + r/100)^n$ ; 1,000 at 10% for 2 years = 1,210.
- A 60% to 72% rise is 12 percentage points but a 20 per cent increase.
- Negative marking of one-third: three wrong answers cancel one correct; 90 correct and 30 wrong on 2-mark questions = 160.
- Share in a ratio = (part / sum of parts) x total; Rs 36,000 in 2 : 3 : 4 = 8,000, 12,000, 16,000.
- Combine  $a : b$  and  $b : c$  by equalising  $b$ ;  $a : b = 3 : 4$  and  $b : c = 2 : 5$  give  $a : b : c = 3 : 4 : 10$ .
- $a : b :: c : d$  means  $ad = bc$ ; mean proportional =  $\sqrt{ab}$ ; third proportional to  $a$ ,  $b = b^2/a$ ; fourth proportional to  $a$ ,  $b$ ,  $c = bc/a$ .
- Duplicate ratio of 2 : 3 is 4 : 9; sub-duplicate ratio of 16 : 25 is 4 : 5.
- Average = sum / count; adding  $k$  to every value raises the average by  $k$ .
- Average of the first  $n$  natural numbers =  $(n + 1)/2$ , so the first 50 average 25.5; any AP averages (first + last)/2.
- Replacement with count unchanged: new value = old value +  $n \times$  (change in average).
- Weighted average =  $(n_1 A_1 + n_2 A_2)/(n_1 + n_2)$ ; 40 at 60 and 60 at 70 give 66, not 65; average

speed over equal distances =  $2xy/(x + y)$ .